

UnoLib documentation

ArduinoISP.pas

version 06/11/2026

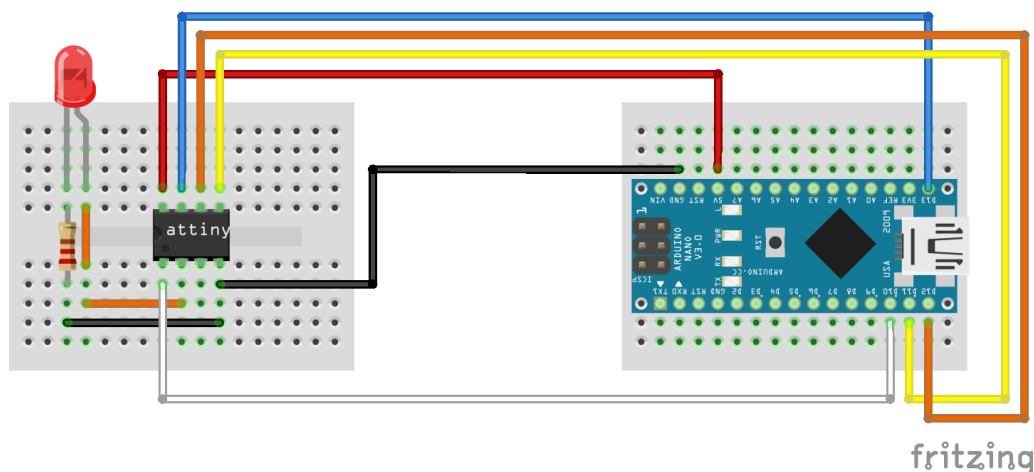
UnoLib includes a Pascal port of the well-known ArduinoISP sketch. It allows an Arduino Nano (or compatible board) to be used as a simple ISP programmer for classic AVR devices such as the ATtiny13A. The example implements the STK500v1 protocol and can be used directly with AVRdude.

Wiring

Connect the Arduino Nano to the target AVR device as follows:

Arduino Nano	ATtiny13A
D10	RESET (pin 1)
D11 (MOSI)	MOSI / PB0 (pin 5)
D12 (MISO)	MISO / PB1 (pin 6)
D13 (SCK)	SCK / PB2 (pin 7)
5V	VCC (pin 8)
GND	GND (pin 4)

The example schematic shows a test LED connected to PB3 (pin 2) through a current-limiting resistor. Do not connect LEDs or other loads directly to ISP signals (MOSI, MISO, SCK, RESET), as they may interfere with programming.



Programming

The ArduinoISP.pas is located in *examples* directory. After uploading ArduinoISP.pas to the Nano, the programmer can be used with AVRdude:

```
avrdude -pt13a -Pcom5 -cstk500v1 -b19200 \  
-Uflash:w:blink.hex:i
```

On Linux:

```
avrdude -pt13a -P/dev/ttyUSB0 -cstk500v1 -b19200 \  
-Uflash:w:blink.hex:i
```

Test LED program:

```
program attiny13_led_ISP;  
  
{IFDEF attiny13}  
  {$Fatal Invalid controller type, expected: attiny13}  
{ENDIF}  
  
procedure HalfSecondDelay;  
label  
  Delay;  
begin  
  asm  
    ldi R25, 234 // 2 clock cycles  
    ldi R24, 96  // 2 clock cycles  
  Delay:  
    sbiw R24,1   // 2 cycles  
    nop          // 1 cycle  
    nop          // 1 cycle  
    nop          // 1 cycle  
    nop          // 1 cycle  
    nop          // 1 cycle  
    nop          // 1 cycle  
    brne Delay  // 2 cycles if jump to label, 1 cycle if not  
    nop          // 1 cycle  
  end;  
end;  
  
const  
  DB3 = %00001000;  
  PB3 = %00001000;  
  
begin  
  DDRB := DDRB or DB3;           // pin 3 as output  
  
  while true do  
    begin  
      PORTB := PORTB or PB3;      // high state on pin 3 -> turn on LED  
      HalfSecondDelay;             // waiting  
      PORTB := PORTB and not PB3;  // low state on pin 3 -> turn off LED  
      HalfSecondDelay;             // waiting  
    end;
```

end.

Notes

- The example is intended for educational purposes and small AVR projects.
- A 100 nF decoupling capacitor between VCC and GND is recommended near the target AVR.
- The target MCU should use the default SPI programming interface (ISP enabled).
- Tested with ATtiny13A and AVRDUDE using the STK500v1 protocol.